

One-Step Synthesis of Cyanated Quinoxaline Regioisomers Enables n-Type Polymers with Record Ambient-Processed Thermoelectric Performance

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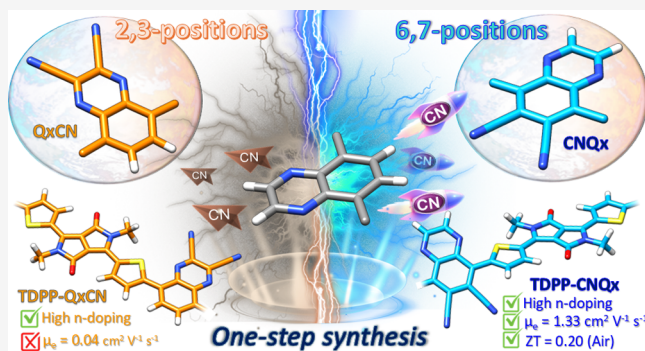
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ABSTRACT: Advancements in organic electronics critically depend on the development of high-performance n-type π -conjugated polymers. However, electron-depleted polymers of this kind exhibiting facile synthesis, excellent processability and good air stability remain rare due to limitations in accessing building blocks meeting these requirements. Here we develop two novel, structurally simple and yet π -extended and highly electron-deficient cyanated building blocks: 2,3-quinoxalinedicarbonitrile (QxCN) and 6,7-quinoxalinedicarbonitrile (CNQx). These units are synthesized in a single-step from low-cost precursors via regioselective cyanation of the weak acceptor quinoxaline. Copolymerization with thiophene-flanked diketopyrrolopyrrole affords two acceptor–acceptor type copolymers TDPP-QxCN and TDPP-CNQx, respectively, exhibiting significantly deeper LUMO levels (~ -3.9 eV) compared to the noncyanated polymer analogue TDPP-Qx (-3.35 eV). In-depth spectroscopic, theoretical, and microstructural analyses reveal that cyanation at the 6,7-positions in TDPP-CNQx suppresses minority carrier effects associated with intramolecular charge transfer in TDPP-QxCN, while promoting excellent conformational adaptability that drives efficient self-assembly into a highly crystalline film with a favorable balance of ordered and disordered domains. This microstructural organization ensures rapid electron transport and efficient dopant diffusion, leading to large field-effect mobilities for TDPP-CNQx (up to $1.3\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$) and unprecedented n-doping efficiency even under air-processed conditions. Consequently, n-doped TDPP-CNQx films exhibit a high n-type electrical conductivity (44.50 S cm^{-1}), excellent power factor ($65.16\text{ }\mu\text{W m}^{-1}\text{ K}^{-2}$), and ultralow thermal conductivity ($0.10\text{ W m}^{-1}\text{ K}^{-1}$), collectively culminating in a record thermoelectric figure of merit ($ZT = 0.20$) for solution/ambient-processed films. This performance also surpasses nearly all vacuum-processed counterparts, positioning CNQx as a highly promising building block for next-generation organic electronics.



INTRODUCTION

Over the past few decades, organic semiconductors have evolved from laboratory curiosities to important materials driving the development of wearable and sustainable electronics.^{1,2} Among them, π -conjugated polymers (CPs) stand out due to their unique combination of mechanical flexibility, lightweight, solution processability, and unparalleled tunability of electronic properties,^{3–5} which have enabled breakthroughs in various devices such as organic thin-film transistors (OTFTs),⁶ organic solar cells,⁷ organic electrochemical transistors,⁸ and organic thermoelectrics (OTEs).⁹ Despite this progress, the field continues to face a fundamental challenge in developing high-performance n-type (electron-transporting) CPs, whose performance still lags far behind their p-type (hole-transporting) counterparts due to the scarcity of n-type polymers that combine high electron affinities, efficient electron transport and strong resistance to oxidative degradation in their n-doped state.^{10,11} While p-type

CPs have achieved broad material diversity and exceptional performance, exemplified by hole mobilities up to $10\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ and electrical conductivities (σ) exceeding 2000 S cm^{-1} in the commercially available polymer poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS in Figure 1a),^{12,13} there are very few unipolar n-type CPs that simultaneously achieve electron mobilities (μ_e 's) above $1\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$ and σ values exceeding 50 S cm^{-1} upon n-doping (Figure 1b).^{14–16} The intrinsically n-doped poly(benzodifurandione) (PBFDO) stands out as the only n-type

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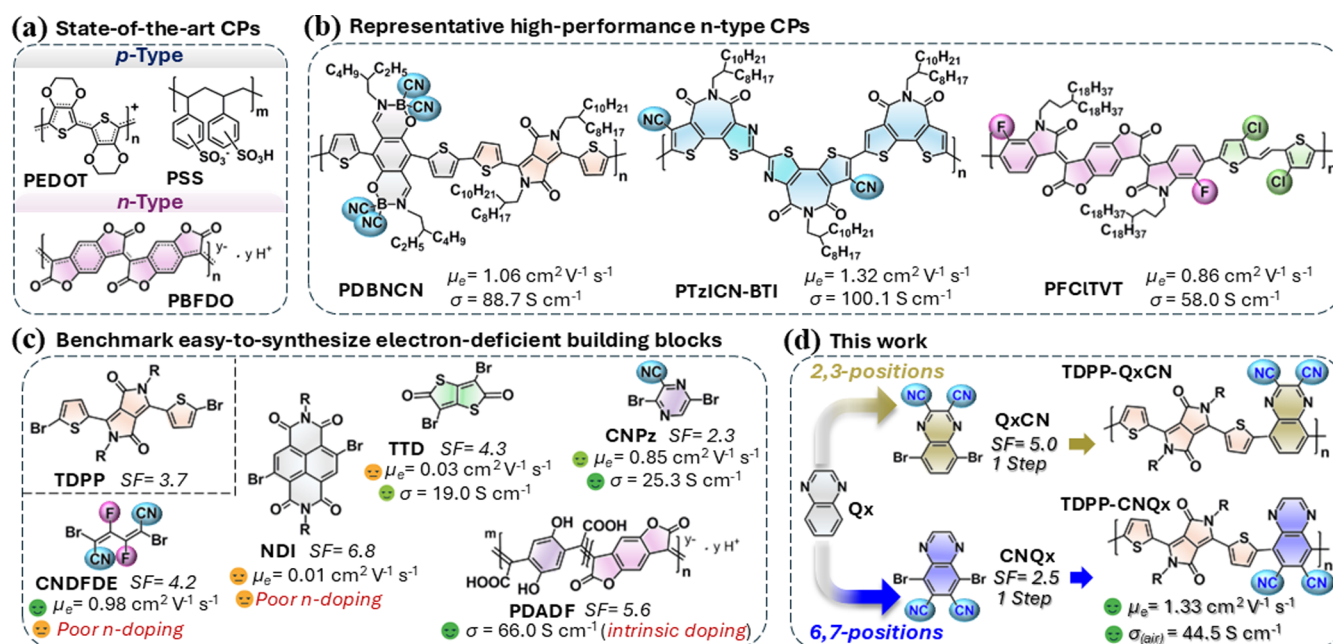


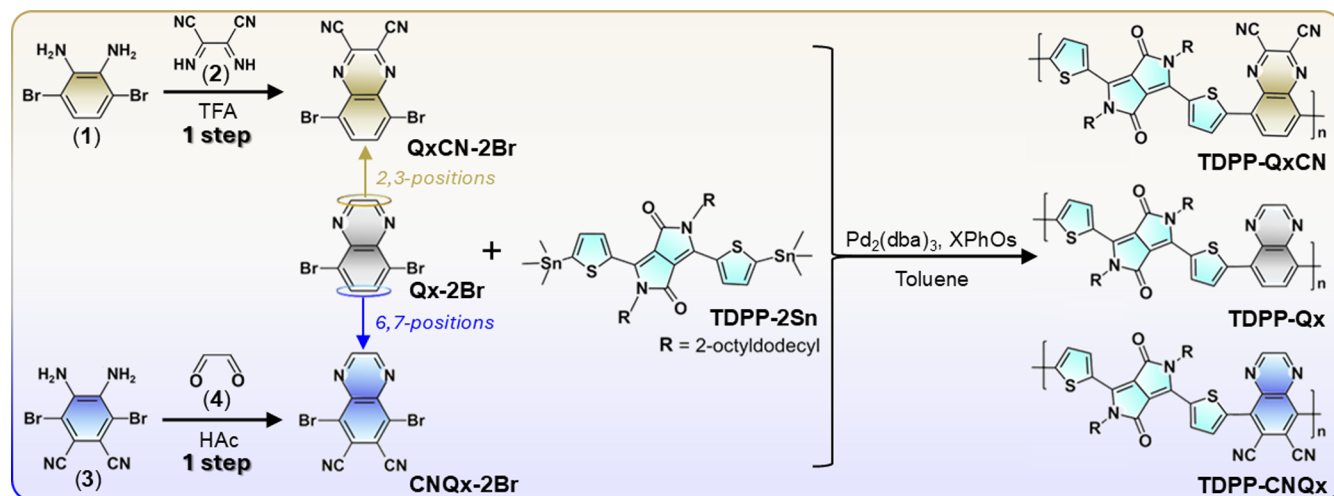
Figure 1. Chemical structures of representative (a) state-of-the-art p- and n-type semiconducting polymers; (b) high-performance n-type CPs simultaneously achieving $\mu_e \approx 1 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and σ exceeding 50 S cm^{-1} upon n-doping; (c) benchmark electron-deficient building blocks readily accessible in one or two synthetic steps. The field-effect μ_e and σ upon n-doping are shown for polymers synthesized by copolymerizing these building blocks with TDPP, as well as for the intrinsically n-doped PDAF polymer based on the BFDO unit. The scalability factor (SF) of these building blocks, calculated from the total number of steps and the reciprocal yield of the synthetic route, is used to quantify their synthetic complexity; (d) cyanated quinoxaline regioisomers developed in this work via one-step synthesis, together with their corresponding TDPP-based polymers.

CP with σ values comparable to PEDOT. Unfortunately, in this polymer it is difficult to control the n-doping level preventing to achieve large field-effect mobility (hall mobility $\sim 0.8\text{--}0.9 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) and thermoelectric response ($ZT \sim 0.26$).¹⁷ Overall, this persistent performance gap hinders the development of complementary logic circuits,¹⁸ thermoelectric modules,¹⁹ and other technologies based on p–n heterojunctions,^{20,21} which require well-matched μ 's and/or σ 's for both hole and electron carriers.

An effective strategy to address this challenge lies in the rational design of new electron-deficient building blocks that enable polymers with good solubility, planar backbones for efficient charge delocalization, and low-lying lowest unoccupied molecular orbital (LUMO) levels (typically below -4.0 eV) to facilitate n-doping and suppress electron trapping by oxygen and moisture.²² However, realizing this multifaceted design is inherently challenging due to the limited availability of strong electron-withdrawing groups (e.g., imides, amides, B $\leftarrow$ N coordination bonds, cyano (CN) units, and halogens), and the dilemma associated with their incorporation in polymer backbone, which typically leads to severe torsions.²³ Among them, cyanation has proven particularly effective for developing high-performance n-type small molecules and CPs due to its strong electron-withdrawing nature, large dipole moment, and commercial availability of CN-containing precursors.^{24–26} For example, Wang et al. developed the DBNCN unit by combining B $\leftarrow$ N and CN functionalization in an anthracene framework, leading to the high-performance polymer PDBNCN with a field-effect μ_e of $1.06 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ and σ of 87.8 S cm^{-1} , placing it among the best-performing n-type CPs reported to date (Figure 1b).¹⁴ In this design, the CN groups are peripherally attached to the boron atom rather than directly to the π -conjugated backbone, resulting in a

modest LUMO stabilization of only 0.05 eV compared to the noncyanated analogue, while minimizing the steric hindrance that could otherwise distort backbone planarity and limit charge transport.²⁷ In fact, this steric issue was directly investigated by Feng et al., who developed the CNI building block by introducing two CN groups into the bithiophene imide (BTI) moiety. While the resulting polymer, PCNI-BTI, exhibited a substantially low LUMO level (-4.10 eV), enabling more efficient n-doping and thereby a notable increase in charge carrier density and σ (23.3 S cm^{-1} vs 0.002 S cm^{-1} for PBTI), PCNI-BTI-based OTFTs showed a much lower μ_e ($0.16 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) than the parent polymer PBTI ($1.19 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) mainly due to the high degree of backbone distortion in PCNI-BTI.²⁸ These studies underscore the delicate trade-off between enhancing electron affinity and preserving structural order, an ongoing challenge requiring deeper insights into structure–property relationships. In addition, many promising electron-deficient building blocks suffer from lengthy multistep syntheses and laborious purification procedures, which increase material costs and create major obstacles to scalability.²⁹ For example, the synthesis of the DBNCN and CNI units involves 7- and 9-steps, respectively, starting from commercially available precursors.^{14,28} This reflects a broader trend in the field, where research efforts have largely prioritized maximizing electronic performance, often at the expense of addressing the equally vital need for cost-effective and scalable synthesis for real-world applications. To date, only a handful of efficient electron-deficient building blocks have been invented as readily accessible via 1- or 2-step syntheses without extensive purification, including naphthalene diimide (NDI),³⁰ thiophene-flanked diketopyrrolopyrrole (TDPP),³¹ thieno[3,2-*b*]thiophene-2,5-dione (TTD),³² 2,3-difluoro-1,4-dicyano-butadiene (CNDFDE),³³ 3,6-dibromo-

Scheme 1. Synthetic Route to the Dibrominated Cyanated Quinoxaline Monomers QxCN-2Br and CNQx-2Br from Commercially Available Starting Materials 1–4, and Their Corresponding A–A Type Copolymers TDPP-QxCN and TDPP-CNQx. Note that the Dibrominated Quinoxaline Monomer Qx-2Br is Commercially Available, and the Corresponding CP (TDPP-Qx) is Also Synthesized for Comparison



pyrazine-2-carbonitrile (CNPz),³⁴ and the DADF unit (Figure 1c).³⁵ Although these building blocks represent important progress toward next-generation n-type CPs, they fall short in achieving both high μ_e and efficient yet controllable n-doping, underscoring an urgent need for new strategies that reconcile high performance with synthetic accessibility (see the scalability factor (SF) in Table S1).

In this work, we aim at addressing these long-standing limitations by introducing a new design strategy that bridges synthetic accessibility, high n-type performance, and ambient processability. Our approach is inspired by the knowledge that, although quinoxaline (Qx)-based polymers have gained growing attention in polymer/perovskite solar cells since the emergence of a low-cost, high-performance donor–acceptor (D–A) copolymer in 2018,³⁶ their potential for n-type applications has remained overlooked. This is particularly surprising given the Qx core’s ready synthetic accessibility, moderate electron-deficiency and multiple substitution positions, which make it a versatile scaffold for further functionalization with strong electron-withdrawing groups.³⁷ Here, we exploit this potential by developing two novel electron-deficient building block regioisomers, 5,8-dibromo-2,3-quinoxalinedicarbonitrile (QxCN-2Br) and 5,8-dibromo-6,7-quinoxalinedicarbonitrile (CNQx-2Br), attained via a regioselective cyanation of the Qx core (Figure 1d). Remarkably, both units are accessible via a single-step synthesis from inexpensive precursors, with CNQx demonstrating a significantly improved scalability factor (SF) compared to the current state-of-the-art, easily synthesized cyanated building blocks (see Table S1 and Note S1). Copolymerization with a thiophene-flanked diketopyrrolopyrrole (TDPP) affords two polymers, TDPP-QxCN and TDPP-CNQx, with the latter exhibiting outstanding n-type charge transport and thermoelectric properties. The key to this performance lies in the strategic cyanation at the Qx 6,7-positions, which simultaneously lowers the LUMO to -3.93 eV, suppresses minority carrier transport, and enhances the polymer’s conformational adaptability to promote optimal thin-film morphology. These features collectively promote electron transport ($\mu_e = 1.33$ cm² V⁻¹ s⁻¹) and excellent ambient n-doping efficiency. As a result,

JLBL-doped TDPP-CNQx achieves a high σ of 44.50 S cm⁻¹ and a record thermoelectric figure of merit ($ZT = 0.20$) at room temperature. This work not only uncovers a vaguely explored structure–property paradigm in n-type CPs, where regioselective building block cyanation profoundly governs conformational adaptability, electronic structure, and solid-state packing of the resulting polymers, but also marks a major advance in the design of next-generation n-type CPs for electronic applications.

RESULTS AND DISCUSSION

Polymer Synthesis and Thermal Properties. The dibrominated quinoxaline monomer Qx-2Br is commercially available and the cyanated monomers, QxCN-2Br and CNQx-2Br, were synthesized in a single step. As illustrated in Scheme 1, QxCN-2Br was obtained from 3,6-dibromo-1,2-benzenediamine (1) via a deamination reaction with 2,3-diiminomaleonitrile (2) in the presence of trifluoroacetic acid (TFA), affording the monomer in a 25% yield. Similarly, CNQx-2Br was synthesized from 4,5-diamino-3,6-dibromophthalonitrile (3) via the reaction with glyoxal (4) in the presence of acetic acid (AcOH), affording the monomer in a 65% yield. The chemical identity and purity of both monomers were confirmed by ¹H NMR, ¹³C NMR and EA (Figures S1–S4 and Table S2). Subsequently, three A–A type polymers TDPP-CNQx, TDPP-QxCN and TDPP-Qx, were synthesized via Stille coupling polymerization of QxCN-2Br and CNQx-2Br monomers as well as the commercially available Qx-2Br, with the well-known electron-deficient distannylated monomer TDPP-2Sn (Scheme 1), respectively. After polymerization, the resulting polymers were purified by Soxhlet extraction with methanol, hexane, and chloroform to remove oligomers and impurities. The final chloroform fractions were collected and reprecipitated into methanol to afford the product polymers for material characterization and device fabrication.

The number-average molecular weights (M_n) and dispersity index ($\mathcal{D}$) values of the corresponding Qx-based polymers were determined by high-temperature gel permeation chromatography (HT-GPC) at 150 °C using trichlorobenzene as the eluent (Figure S5). TDPP-Qx, TDPP-CNQx and

Table 1. Summary of the Molecular Weights, Optical and Electrochemical Properties of the Quinoxaline-Based Polymers

polymer	M_n (kDa) ^a	$\bar{D}$ ^a	$\lambda_{\text{sol max}}$ (nm) ^b	$\lambda_{\text{film max}}$ (nm) ^c	$\lambda_{\text{film onset}}$ (nm) ^d	$E_{\text{opt g}}$ (eV) ^e	E_{LUMO} (eV) ^f	E_{HOMO} (eV) ^f
TDPP-Qx	70.4	2.0	874	906	1015	1.22	−3.35	−5.19
TDPP-CNQx	153.7	2.8	894	934	1120	1.11	−3.93	−5.59
TDPP-QxCN	15.6	3.2	976	1040	1242	0.99	−3.91	−5.47

^aDetermined by GPC using trichlorobenzene as the eluent at 150 °C. ^bFrom dilute chloroform solutions (10^{-5} M). ^cFrom as-cast films. ^dEstimated from the optical absorption onset of polymer films. ^eOptical bandgap ($E_{\text{opt g}}$) derived from $\lambda_{\text{film onset}}$ using the equation: $E_{\text{opt g}} = 1240/\lambda_{\text{film onset}}$. ^fDerived from cyclic voltammetry (CV) measurements using the equations: $E_{\text{LUMO}} = -(4.80 + E_{\text{onset red}})$ eV and $E_{\text{HOMO}} = -(4.80 + E_{\text{onset ox}})$ eV. The reduction ($E_{\text{onset red}}$) and oxidation ($E_{\text{onset ox}}$) potentials were electrochemically determined versus the Fc/Fc⁺ redox couple.

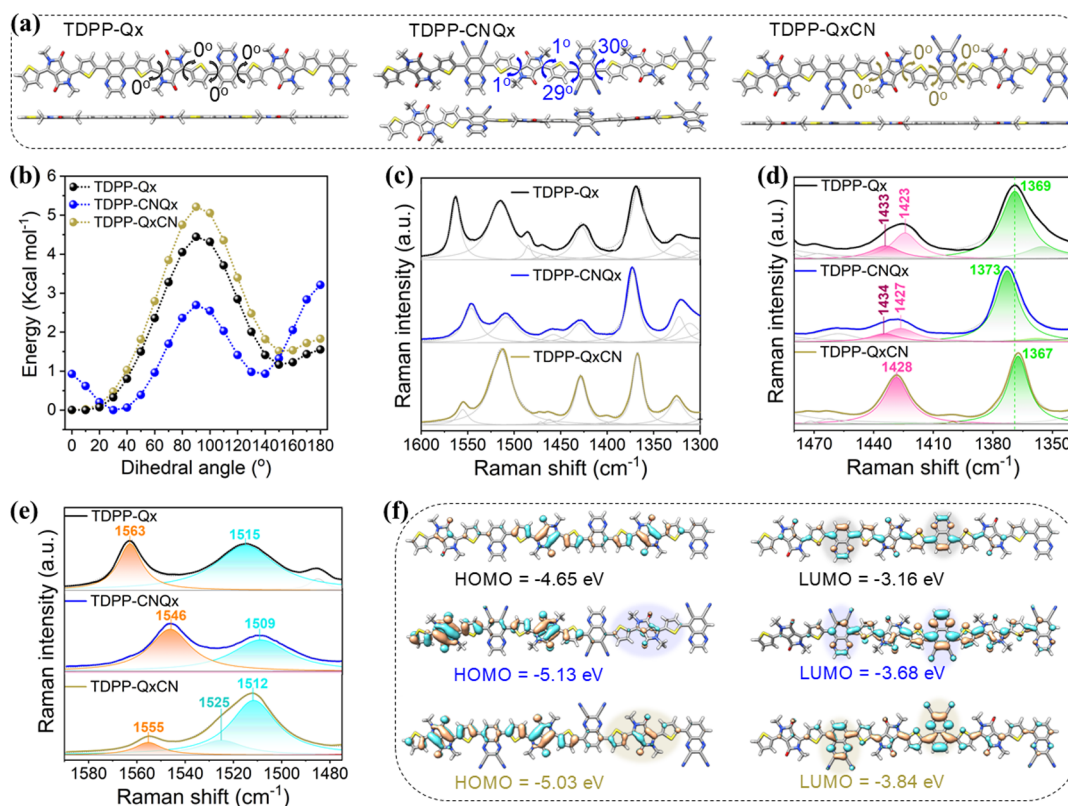


Figure 2. (a) DFT-optimized molecular structures of trimeric models of polymers **TDPP-Qx**, **TDPP-CNQx** and **TDPP-QxCN**. The inter-ring dihedral angles between nonfused moieties are also shown. (b) Potential energy surface for the dihedral angle connecting both Qx-based and TDPP co-units. (c) Raman spectra of polymer films under 532 nm excitation, with a single-component band deconvolution based on Lorentzian fitting. (d) Optimal deconvolution which best fits the representative Raman bands between 1480 and 1340 cm⁻¹. Band deconvolutions are shown using a Lorentzian fitting and indicate the evolution of more conjugated (pink) and less conjugated (purple) phases with regioselective cyanation. (e) Optimal deconvolution which best fits the representative Raman bands between 1590 and 1470 cm⁻¹. (f) DFT-calculated HOMO and LUMO wave functions of trimeric models of the three Qx-based polymers.

TDPP-QxCN exhibited M_n values of 70.4, 153.7, and 15.6 kDa, as well as $\bar{D}$ values of 2.0, 2.8, and 3.2 (Table 1), respectively. Notably, the M_n of **TDPP-CNQx** is considerably higher than that of its counterpart **TDPP-QxCN**, owing to the latter's excessive aggregation and reduced solubility. This results in polymer precipitation once M_n exceeds a threshold value, even after an exceptionally short polymerization time of just ~20 min. Interestingly, thermogravimetric analysis (TGA) revealed excellent thermal stabilities for these polymers, with **TDPP-Qx**, **TDPP-CNQx** and **TDPP-QxCN** exhibiting 5% mass decomposition temperatures exceeding 400 °C (Figure S6). Additionally, differential scanning calorimetry (DSC) analysis unveiled no apparent phase transitions for any of the polymers within the temperature range of 25–300 °C (Figure S7). The demonstrated thermal stability is advantageous for device fabrication and enables morphology optimization across a broad processing temperature window.

Key Structural Characteristics of the Polymers. To gain insight into the effects of cyanation in the structural and electronic properties of the newly synthesized Qx-based A–A type polymers, density functional theory (DFT)-based calculations were performed on trimeric models of these polymers at the B3LYP-D3/6–31G** level using the Gaussian 16 software. As shown in Figure 2a, theoretical modeling predicts a completely planar polymer backbone for **TDPP-Qx**, presumably enabled by effective intrachain noncovalent N⋯S interactions between the lone electron pairs of nitrogen atoms in the Qx units and the sulfur atoms of adjacent thiophene moieties.^{38,39} In fact, the calculated N⋯S distances (2.77 Å) were found to be far shorter than the sum of their van der Waals radii (3.35 Å), indicative of the presence of through-space intramolecular noncovalent interactions that promote a conformationally locked polymer backbone. Interestingly, introducing CN groups at the 2,3-positions of the Qx unit in

TDPP-QxCN not only preserves the structural coplanarity of the π -conjugated system but also enhances backbone rigidity. This is evidenced by the higher energy barrier calculated for molecular torsion between the cyanated acceptor unit and adjacent thiophene moieties compared to that of the unfunctionalized polymer (**Figure 2b**). The effect arises primarily from the increased electrophilicity of the nitrogen atoms of the pyrazine ring, where the electron-withdrawing CN groups deplete the nitrogen atom electron density, thereby strengthening their ability to interact with the electron-rich sulfur atoms in the adjacent thiophene.^{34,40} The enhanced polarization at the N(Qx)⋯S contacts is further supported by DFT-calculated Mulliken atomic charges, which reveal more negative values for the nitrogen atoms in the Qx units ($-0.55 e$ vs $-0.50 e$) and more positive values for the sulfur atoms ($+0.34 e$ vs $+0.32 e$) upon cyanation in **TDPP-QxCN**. On the other hand, a markedly different scenario emerges upon cyanation at the 6,7-positions in the **TDPP-CNQx** analogue. Specifically, the CN groups lead to moderate steric hindrance with the neighboring hydrogen atoms of adjacent thiophene moieties, resulting in a moderately twisted backbone with a dihedral angle of about 30° between the two acceptor counts. Apparently, such a more twisted backbone may hinder tight intermolecular π - π stacking, potentially reducing charge carrier transport efficiency compared to **TDPP-Qx** and **TDPP-CNQx**, which a priori feature a highly planar skeleton.⁴¹ However, it is also important to emphasize here that the calculated potential energy surface of **TDPP-CNQx** reveals significant conformational flexibility about the C-C bond connecting the CNQx unit and adjacent thienyl rings, with a small energy barrier of only ~ 1 kcal mol⁻¹ for adopting the coplanar conformation (**Figure 2b**). Therefore, given this high rotational flexibility and considering that these calculations are based on isolated molecules in vacuum, it seems reasonable to assume that intermolecular interactions in condensed states may promote backbone planarization.

To shed light on the polymer conformation in the solid state, Raman spectra of the polymer films were recorded under a 532 nm excitation. As shown in **Figure 2c**, the spectra display a complex vibrational profile that can be deconvoluted into several peaks associated with the main C-C/C=C stretching modes of the π -conjugated backbone. However, it should be noted that a single-component Lorentzian fitting fails to adequately resolve the bands around 1430 and 1370 cm⁻¹, which arise from modes mostly localized on the pendant thiophene rings and adjacent single C-C bonds, with the latter mode extending into the DPP unit (see eigenvectors in **Figures S8–S10**).^{42,43} Since skeletal disruptions affecting charge transport properties are closely associated to these rotationally flexible modes,^{44–46} and given their pronounced sensitivity to dihedral angle variations between monomeric units (as theoretically demonstrated in **Figure S11**), the position and shape of these bands serve as valuable indicators of intramolecular conformational order in solid state. In this context, the band observed around 1430 cm⁻¹ appears broad in **TDPP-Qx** and becomes progressively sharper in **TDPP-CNQx** and **TDPP-QxCN**, suggesting the coexistence of multiple phases with different degrees of intrachain conformational order.^{47,48} To further investigate this behavior, we performed a multicomponent peak deconvolution, which revealed optimal fits when two overlapping peaks were assumed for both **TDPP-Qx** and **TDPP-CNQx**. These are presumably attributed to the coexistence of an ordered phase with a band centered at ~ 1425

cm⁻¹ (highlighted in pink in **Figure 2d**), and a disordered phase with a band around 1433 cm⁻¹ (highlighted in purple in **Figure 2d**). By contrast, a single-component fitting was sufficient for **TDPP-QxCN**, indicative of an exclusively ordered phase with a band centered at 1428 cm⁻¹. These findings imply that CN-substitution of the Qx unit enhances supramolecular organization within the polymer, with substitution at the 2,3-positions (**TDPP-QxCN**) being particularly effective in promoting backbone planarity by further restricting the conformational freedom of the building blocks.

This effect becomes evident from the comparison of the Raman spectra recorded under 473 and 785 nm excitations. Note that alternatively to the 473 and 532 nm lasers, excitation at 785 nm is resonant with the higher-energy region of the electronic absorption band, selectively enhancing signals from the more ordered phase (see **Figure S12**). As illustrated in **Figure S13**, the Raman band at 1428 cm⁻¹ in **TDPP-QxCN** remains symmetrical independently of the wavelength used, confirming the presence of a single ordered phase within the nanodomains due to its superior backbone rigidity. In contrast, **TDPP-CNQx** consistently displays features of both ordered and disordered phases under all excitation conditions, highlighting its greater conformational flexibility. Notably, a nearly balanced area ratio between the ordered and disordered phases ($A_{\text{ORD}}/A_{\text{DIS}}$) of 0.99 was found even under non-resonant condition at 473 nm excitation, which increases significantly to 1.77 upon 785 nm excitation. This result suggests that, despite the coexistence of phases, the ordered one dominates in solid state likely due to efficient self-assembly into ordered domains.

Interestingly, the imprint of this structural evolution is also reflected in the DPP moiety, where the Raman band at 1367 cm⁻¹ (highlighted in green in **Figure 2d**) sharpens progressively from the parent **TDPP-Qx** to **TDPP-CNQx** and ultimately to **TDPP-QxCN**, accompanied by slight shifts in frequency. Specifically, **TDPP-QxCN** exhibits a small downshift of 2 cm⁻¹ relative to **TDPP-Qx**, consistent with a higher π -conjugation degree conferred by its more conformationally locked backbones. On the other hand, a small upshift of 4 cm⁻¹ is noted for its analogue **TDPP-CNQx**, likely arising from electronic structure changes in the DPP moiety driven by minor backbone distortions (see **Figure S14**). However, based on the concurrent band sharpening and the minor magnitude of this upshift in **TDPP-CNQx**, far below the 16–20 cm⁻¹ shifts predicted by gas-phase DFT calculations for large distortion angles, it seems reasonable to conclude that **TDPP-CNQx** can achieve effective backbone planarization in film state.

Beyond structural effects, we also noted that the different substitution patterns modulate the electron-withdrawing strength of the Qx units, directly impacting the intramolecular charge transfer (ICT) character between the TDPP and Qx units. This is clearly manifested in the notable downshifts and relative intensity variations of the Raman bands around 1550 and 1510 cm⁻¹, primarily associated with Qx- and TDPP-localized modes, respectively (highlighted in orange and cyan in **Figure 2e**). A more detailed analysis of these effects is provided in **Note S2**, where it is shown that the planar backbone of **TDPP-QxCN** enables stronger electronic coupling across the TDPP unit, leading to suppressed Qx-localized vibrations and enhanced TDPP signatures (see **Figures S15 and S16**). Consistent with these spectroscopic observations, frontier molecular orbital (FMOs) analysis of the

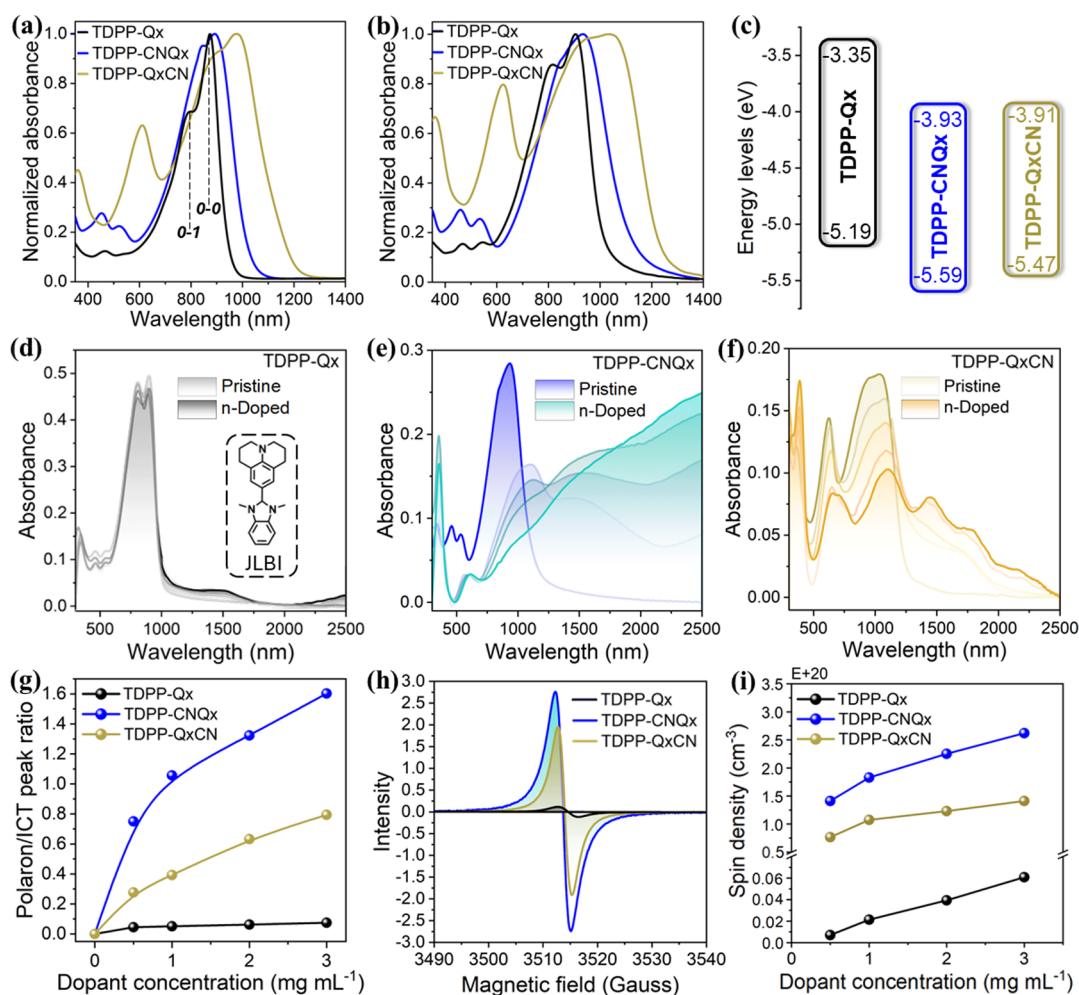


Figure 3. Normalized UV-vis absorption spectra of polymers **TDPP-Qx**, **TDPP-CNQx** and **TDPP-QxCN**, measured in (a) 10^{-5} M chloroform solutions and (b) as thin-films. (c) FMO energy levels of polymer films obtained from cyclic voltammetry measurements. UV-vis-NIR absorption spectra of the pristine and JLBI-doped films of (d) **TDPP-Qx**, (e) **TDPP-CNQx** and (f) **TDPP-QxCN** upon increasing dopant concentration in the range of 0.5–3 mg mL⁻¹. Doped films were measured under inert conditions. (g) Doping levels calculated from polaron/ICT absorption peak ratios for n-doped films of the newly synthesized polymers. The intensity of the polaron peaks was measured at the following peak positions: 1395 nm for **TDPP-Qx**, 1486 nm for **TDPP-CNQx**, and 1446 nm for **TDPP-QxCN**. (h) ESR signals of pristine and n-doped polymer films at a JLBI concentration of 3 mg mL⁻¹. (i) Spin densities extracted from the ESR signals as a function of the dopant loading.

quinoxaline-based and TDPP building blocks further rationalizes the stronger ICT character of **TDPP-QxCN** compared to **TDPP-CNQx** (Figure S17). As expected, cyanation of the Qx core stabilizes both the highest occupied molecular orbital (HOMO) and LUMO, with QxCN exhibiting a lower LUMO (−3.29 eV vs −3.16 eV for CNQx) and a slightly higher HOMO (−7.65 eV vs −7.73 eV), indicative of its stronger electron-withdrawing nature. Upon coupling with TDPP, these electronic trends are preserved (Figure 2f): **TDPP-QxCN** displays a more stabilized LUMO (−3.84 eV) and a slightly higher HOMO (−5.03 eV) than **TDPP-CNQx** (LUMO: −3.68 eV, HOMO: −5.13 eV). As a result, **TDPP-QxCN** shows a narrower energy gap (1.19 vs 1.45 eV), which likely favors charge transfer. Orbital visualization supports this interpretation, revealing stronger spatial separation in **TDPP-QxCN**, with the LUMO largely localized on the QxCN acceptor, whereas the LUMOs of **TDPP-Qx** and **TDPP-CNQx** are more delocalized along the backbone, as expected for A–A type copolymers. Importantly, **TDPP-QxCN** also exhibits enhanced HOMO delocalization across adjacent TDPP units, indicative of a more donor-like character in

agreement with its higher HOMO energy. Collectively, these features demonstrate that the synergistic effects of backbone planarization and the stronger electron-withdrawing ability of QxCN enhance ICT in **TDPP-QxCN** by rendering the TDPP unit relatively more electron-donating with respect to the electron-deficient QxCN core.^{14,49}

Polymer Optoelectronic Properties. To gain additional understanding into how the substitution pattern affects the electronic properties of the polymers, ultraviolet–visible–near-infrared (UV-vis-NIR) absorption spectra of the polymers both in chloroform solution and as thin films were recorded, with the absorption data summarized in Table 1. As seen in Figure 3a, all polymers display characteristic dual-peak absorption profiles, featuring a weak band in the 450–650 nm range and a strong, shoulder-shaped band typical of TDPP-based material systems that extends broadly into the near-infrared region (700–1200 nm).^{37,50,51} Time-dependent (TD)-DFT calculations attribute the high-energy band to local π – π^* electronic transition, whereas the dominant low-energy band ($S_0 \rightarrow S_1$) corresponds to an ICT transition from the HOMO, primarily localized on the TDPP unit, to the

LUMO, which is mainly centered on the Qx unit but also extends over the flanking thiophene rings (see Figure S18). Notably, it was found that the optoelectronic properties of the parent TDPP-Qx polymer can be profoundly tuned through regioselective cyanation. Functionalization at the 6,7-positions in TDPP-CNQx induces a moderate bathochromic shift of 20 nm in the solution-phase absorption maximum ($\lambda_{\text{sol max}}$), which shifts from 874 nm (TDPP-Qx) to 894 nm (TDPP-CNQx) while extending the absorption onset beyond 1000 nm. Interestingly, a substantially greater shift is observed for the 2,3-functionalized TDPP-QxCN, which shows a $\lambda_{\text{sol max}}$ centered at 976 nm, representing a remarkable redshift of 102 nm relative to the parent polymer TDPP-Qx and 82 nm compared to its TDPP-CNQx isomer. This marked shift underscores the enhanced D–A character of TDPP-QxCN, driven by a stronger ICT character between the TDPP moiety and the electron-deficient QxCN unit. Closer scrutiny reveals that these copolymers display structured vibronic features in the low-energy absorption band, with two obvious 0–0 and 0–1 peaks. Notably, cyanation increases the relative intensity of the 0–1 peak, indicative of enhanced interchain interactions. This effect is especially pronounced in TDPP-QxCN, which retains partial aggregated features even at low concentrations (Figure S19) and elevated temperatures (Figure S20), attributable to its highly planar and conformationally locked backbone promoting stronger aggregation compared to TDPP-CNQx (see Note S3). This trend aligns with the notable redshifts observed upon film formation, where the cyanated polymers display larger bathochromic shifts in their absorption maxima as a consequence of enhanced inter- and intramolecular interactions (Figure 3b). Specifically, TDPP-Qx, TDPP-CNQx and TDPP-QxCN show progressive redshifts of 32, 40, and 64 nm, respectively, resulting in thin-film absorption maxima ($\lambda_{\text{film max}}$) of 906, 934, and 1040 nm.

Based on the absorption onsets of polymer films, the estimated optical bandgaps ($E_{\text{opt g}}$) progressively decrease from 1.22 eV for TDPP-Qx to 1.11 eV for TDPP-CNQx, reaching a minimum value of 0.99 eV for TDPP-QxCN. This trend further emphasizes the significant influence of the substitution patterns on the optoelectronic characteristics of the resulting polymers. To further elucidate the impact of cyanation on FMOs, the HOMO and LUMO energy levels were estimated by cyclic voltammetry (CV), with the corresponding data summarized in Table 1. As shown in Figure S21, all three polymers display distinctive reduction peaks and relatively weak oxidation peaks. The incorporation of electron-withdrawing CN groups onto the Qx unit lowers the reduction potential while systematically increasing the oxidation potential, indicating enhanced electron affinity and suppressed oxidative character. According to their reduction/oxidation onsets, the LUMO/HOMO energy levels of TDPP-Qx, TDPP-CNQx and TDPP-QxCN were calculated to be $-3.35/-5.19$, $-3.93/-5.59$, and $-3.91/-5.47$ eV, respectively, resulting in electrochemical bandgaps of 1.84, 1.66, and 1.56 eV (Figure 3c). These values are in good agreement with the trends found for the estimated $E_{\text{opt g}}$ values, DFT calculations and ultraviolet photoelectron spectroscopy (UPS) measurements (Figure S22). Notably, the deepest-lying FMOs are observed for TDPP-CNQx, which is not only advantageous for promoting electron injection while suppressing hole accumulation in OTFT devices but also makes it particularly suitable for efficient n-doping (vide infra).

n-Doping Characteristics. To uncover the n-dopability of these polymers, UV–vis–NIR absorption spectroscopy was first employed to monitor n-polaron formation in polymer films as the dopant concentration increases from 0.5 to 3 mg mL⁻¹ (Figures 3d–f). The molecular dopant JLBI (structure shown in the inset of Figure 3d) was chosen for its excellent solution processability, strong reducing capacity, and minimal microstructure disruption during n-doping.^{45,52} All polymers were n-doped using the sequential doping protocol assisted by the transition metal catalyzed doping technique (see Note S4).⁵³ As anticipated, a clear trend was observed wherein doping levels progressively increase as the polymer LUMO level becomes deeper. Specifically, while the CN-free TDPP-Qx polymer (E_{LUMO} : -3.35 eV) remains undoped even at the highest JLBI loading of 3 mg mL⁻¹ (Figure 3d), as evidenced by the nearly unchanged absorption profile, the cyanated Qx-based polymers (E_{LUMO} : ~ -3.9 eV) are considerably n-doped even at a reduced JLBI loading of 0.5 mg mL⁻¹ (Figure 3e,f). This is reflected in the progressive bleaching of the neutral absorption bands located in the 600–1200 nm range, accompanied by the emergence of two prominent sub-bandgap polaronic absorption bands in the visible/near-infrared region: a P1 absorption band localized around 1500 nm (ascribed to the HOMO $\rightarrow$ SOMO electronic transition), and a low-energy P2 absorption band localized in the near-infrared region (ascribed to the SOMO $\rightarrow$ LUMO electronic transition), as further analyzed in Figure S23. Interestingly, despite their nearly identical LUMO levels, noticeable differences in the n-polaron formation are observed between the cyanated polymers. To assess this aspect, the doping efficiency was evaluated using the polaron/ICT peak intensity ratio, where stronger polaron and weaker ICT absorption indicate a higher population of charged polymer chains. Accordingly, TDPP-CNQx exhibited more than twice the polaron/ICT ratio of TDPP-QxCN at identical JLBI concentrations, qualitatively indicating superior doping efficiency for the former (Figure 3g). This trend was further corroborated by electron spin resonance (ESR) measurements, which provide a quantitative assessment of radical concentrations. As expected, no ESR signal was detected in the pristine films, while the doped films show increasing signal intensity with higher dopant loading, consistent with a growing population of charge carriers (see Figure S24). Interestingly, JLBI-doped films of TDPP-CNQx exhibited markedly stronger ESR signals compared to TDPP-QxCN, thereby reflecting more efficient charge generation in these polymer films (Figures 3h,i). Specifically, spin densities of 2.62×10^{20} and 1.39×10^{20} cm⁻³ are measured for doped TDPP-CNQx and TDPP-QxCN films at a JLBI concentration of 3 mg mL⁻¹, respectively, both significantly higher than that of the noncyanated TDPP-Qx reference (6.08×10^{18} cm⁻³). These findings demonstrate that, despite their comparable FMO energies, TDPP-CNQx and TDPP-QxCN exhibit markedly different n-doping efficiencies. This intriguing disparity points to the critical role of regioselective cyanation in modulating polymer-dopant miscibility, a key factor that will be explored in detail below.

Charge Transport Properties. To probe the effect of regiochemical cyanation on the charge transport properties of the Qx-based polymers, OTFTs were fabricated using a top-gate/bottom-contact (TG/BC) device architecture of glass/Au/semiconductor/CYTOP/Al, where the perfluoro(1-butenyl vinyl ether) polymer (CYTOP) was used as the gate dielectric layer (see Supporting Information for detailed

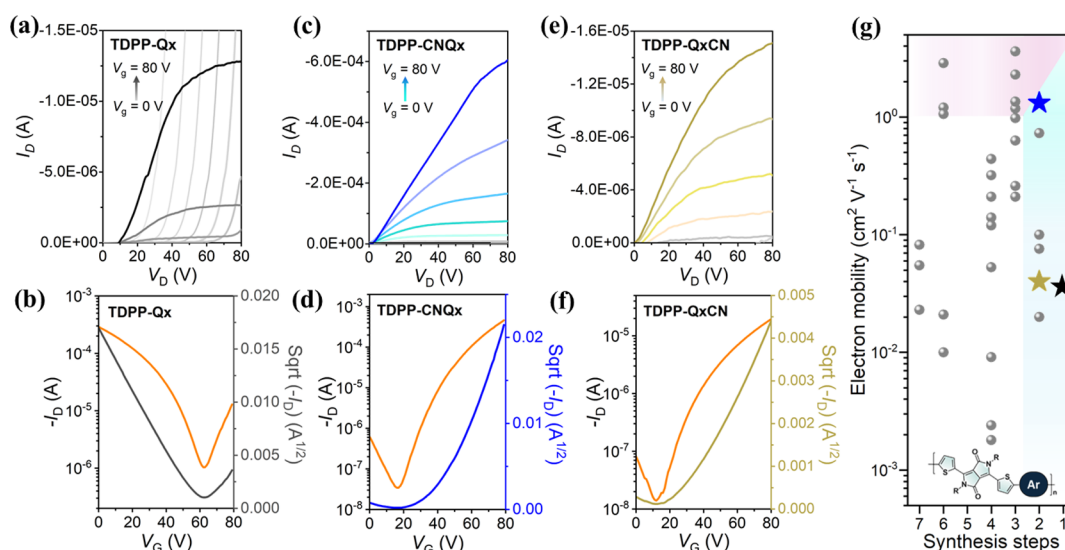


Figure 4. (a,c,e) Output and (b,d,f) transfer characteristics of OTFTs based on (a,b) TDPP-Qx, (c,d) TDPP-CNQx and (e,f) TDPP-QxCN. In the output plots, the gate voltage (V_G) is swept from 0 to 80 V with a 10 V interval. All curves were obtained from devices with a channel length of 20 μm and a channel width of 5 mm. (g) Comparison of the electron mobilities measured in this work (black, blue and gold stars for TDPP-Qx, TDPP-CNQx and TDPP-QxCN, respectively) with those reported in literature for high-performance n-type TDPP-based polymers (see Table S3 and Figure S29 for details).

Table 2. Top-Gate/Bottom-Contact OTFT and Thermoelectric Performance Parameters of the Newly Synthesized Qx-Based n-Type Polymers

polymer	OTFT			
	μ_{e} (cm ² V ^{−1} s ^{−1}) ^a	V_{th} (V) ^b	$I_{\text{on}}/I_{\text{off}}$ ^b	
TDPP-Qx	0.03 (0.02 ± 0.005)	64.1 ± 1.2	208.2 ± 13.3	
TDPP-CNQx	1.33 (1.22 ± 0.06)	37.2 ± 4.0	(5 ± 1.2) × 10 ⁴	
TDPP-QxCN	0.04 (0.03 ± 0.006)	20.8 ± 0.9	(8 ± 2.4) × 10 ³	
polymer	OTE ^c			
	σ (S cm ^{−1})	ISI (μV K ^{−1})	PF (μW m ^{−1} K ^{−2})	ZT
TDPP-CNQx (vacuum)	73.75 (73.47 ± 0.13)	98.23 (97.00 ± 1.47)	71.16 (69.12 ± 1.69)	0.22
TDPP-CNQx (air)	44.50 (43.42 ± 0.55)	121.01 (118.87 ± 2.44)	65.16 (61.35 ± 2.45)	0.20

^aMaximum electron mobilities, with average values (in parentheses) calculated from five independent devices in the saturation regime. ^bAverage values obtained from five independent devices. ^cMaximum OTE device performance parameters at the optimal JLBI concentration of 1 mg mL⁻¹, with average values (in parentheses) from five independent devices.

fabrication and characterization procedures). Note that TDPP-based polymers typically exhibit ambipolar or even preferentially p-type transport characteristics, primarily due to the presence of two electron-donating thiophene moieties within the TDPP unit and the delocalization of the amide nitrogen lone pairs in the polymer backbone.^{54,55} Consistent with this behavior, the reference polymer TDPP-Qx, without CN groups, exhibits a clear ambipolar transport (Figures 4a,b and S27). However, a markedly different behavior emerges upon cyanation, with both polymers TDPP-CNQx and TDPP-QxCN exhibiting intrinsic unipolar n-type transport (see Table 2), characterized by a low off-current ($\sim 10^{-8}$ A) and a moderate on/off current ratio ($I_{on}/I_{off} \approx 10^4$), presumably due to the effective stabilization of both LUMO and HOMO levels enabled by the electron-withdrawing CN groups. Importantly, transfer curve analysis revealed that CN substitution at the 6,7-positions of the Qx unit effectively enhances the electron-transporting characteristics of the resulting polymer TDPP-CNQx, delivering an excellent μ_e of 1.33 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$ and a V_{th} of 37 V (Figure 4c,d). This mobility value is among the highest reported to date for a TDPP-based n-type copolymer (Figure 4g and Table S3),^{6,14,55} further

underscoring the great potential of CNQx as a structurally simple yet powerful electron-deficient unit for the development of high-performance n-type polymers. In stark contrast, cyanation in the TDPP-QxCN analogue led to a substantial decline in electron transport performance, with μ_e dropping to 0.04 $\text{cm}^2 \text{V}^{-1} \text{s}^{-1}$ (Figure 4e,f). This deterioration is primarily attributed to its poor thin film morphology (vide infra).

In addition, space-charge-limited current (SCLC) analysis revealed a significantly higher trap-filled limit voltage (V_{TFL}) for electron-only devices based on TDPP-QxCN (0.69 V) compared with TDPP-CNQx (0.40 V), indicating a markedly higher trap density in the former likely arising from its stronger ICT character (Figure S28).²⁸ This increased trap density promotes electron trapping and is consistent with the inferior charge-transport performance observed for TDPP-QxCN.

Thermoelectric Performance. Recognizing that electrical conductivity ($\sigma = \mu ne$) is governed by both charge carrier mobility and carrier concentration (n), the large μ_e and n-doping efficiency of TDPP-CNQx motivated a comprehensive investigation of its electrical and thermoelectric properties. To this end, OTE devices based on JLBI-doped TDPP-CNQx films were initially fabricated and characterized in a N₂-filled

glovebox (see Supporting Information for device fabrication details). As illustrated in Figure 5a, the σ of TDPP-CNQx

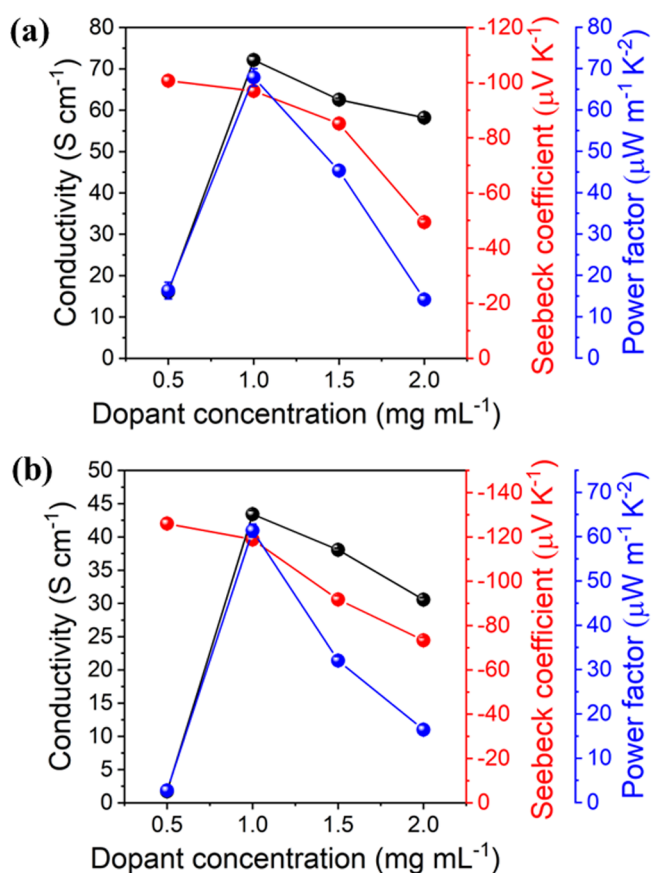


Figure 5. Average electrical conductivities (black), Seebeck coefficients (red), and power factors (blue) of TDPP-CNQx films doped with different concentrations of JLBI under (a) vacuum and (b) ambient conditions.

films increases dramatically upon doping, reaching a maximum value of 73.75 S cm^{-1} at a JLBI concentration of 1 mg mL^{-1} (see Table 2). Further increasing the dopant loading leads to a gradual decline of σ to 58.16 S cm^{-1} for a 2 mg mL^{-1} concentration, a trend commonly attributed to the interplay of carrier–carrier repulsion and dopant-induced microstructural disorder in heavily doped films.⁵⁶ In comparison, the maximum σ 's of optimally doped TDPP-Qx and TDPP-QxCN films are much lower (0.47 and 0.03 S cm^{-1} for JLBI concentrations of 3 and 1 mg mL^{-1} , respectively), consistent with the poor n-doping affinity of TDPP-Qx and the limited electron transport of TDPP-QxCN (see Figure S30). To further probe these differences, charge carrier mobilities were calculated from the experimentally determined σ and n at the respective optimal JLBI loadings using the equation $\sigma = \mu n e$ (where e is the elementary charge). The resulting mobilities of 0.48 , 2.34 , and $0.002 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ for TDPP-Qx, TDPP-CNQx, and TDPP-QxCN (Table S4), respectively, further underscore the decisive role of regioselective cyanation of the quinoxaline core in modulating charge transport.

To quantify thermoelectric performance, Seebeck coefficient (S) values were extracted by measuring the thermovoltage generated across the n-doped films under applied temperature gradients from 0 to 2.8 K in 0.7 K increments (Figures S31–S34). As expected for n-type unipolar systems, all doped films

exhibit negative S values, indicating electron-dominated transport. Increasing the JLBI concentration from 0.5 to 2 mg mL^{-1} leads to a gradual $|S|$ decline from 102.31 to $49.37 \text{ } \mu\text{V K}^{-1}$, reflecting the typical inverse correlation between S and n . Accordingly, higher σ is accompanied by a lower $|S|$ and vice versa, with the optimal balance achieved at a 1 mg mL^{-1} JLBI load. At this concentration, the combination of maximum σ and a still relatively high S ($-98.23 \text{ } \mu\text{V K}^{-1}$) yields the highest power factor (PF = σS^2) of $71.16 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$. Remarkably, TDPP-CNQx maintains a robust n-doped state even when processed under ambient conditions, with films processed in air exhibiting comparable polaron/ICT absorption features to those processed in inert atmosphere and a moderate electron transport stability (see Figure S26). Notably, TDPP-CNQx films processed entirely in ambient retain an excellent σ of 44.50 S cm^{-1} , S of $-121.01 \text{ } \mu\text{V K}^{-1}$, and a PF of $65.16 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$ at the optimal JLBI concentration of 1 mg mL^{-1} (Figure 5b), making it a rare example of an air-processed n-type polymer with substantial thermoelectric performance. It should be noted that although the σ - S trade-off trend mirrors that of vacuum-processed films, slightly higher $|S|$ values are observed for processing in ambient due to partially suppressed carrier concentration. Such ambient-processability is particularly relevant for wearable thermoelectric applications, where sustaining a temperature gradient between the human body and the surrounding environment is critical for efficient energy harvesting.^{57,58} However, ensuring low thermal conductivity (κ) for processing in ambient conditions remains essential for their viability. To assess this aspect, the thermal conductivity of TDPP-CNQx films was measured at room temperature using the transient hot-wire method,^{59,60} enabling full determination of the dimensionless figure of merit $ZT = \sigma S^2 T / \kappa$ (where T is the absolute temperature). In this context, the pristine TDPP-CNQx film exhibits a remarkably low κ of $0.121 \pm 0.021 \text{ W m}^{-1} \text{ K}^{-1}$ at $30 \text{ } ^\circ\text{C}$, ranking among the lowest values reported for n-type organic semiconductors, which typically exceed $0.2 \text{ W m}^{-1} \text{ K}^{-1}$.^{61–63} Upon doping with 1 mg mL^{-1} of JLBI, κ further decreases to $0.110 \pm 0.011 \text{ W m}^{-1} \text{ K}^{-1}$ due to the enhanced phonon scattering by free carriers and dopant-induced molecular packing disruptions.^{64–66} This exceptional thermal insulation, combined with the high PF of $65.16 \text{ } \mu\text{W m}^{-1} \text{ K}^{-2}$, results in a record-breaking ZT value of 0.20 at room temperature. To our knowledge, this represents the highest ZT reported to date for air-processed n-type organic thermoelectrics,^{9,15,67–70} surpassing nearly all vacuum-processed systems except one.⁷¹ Importantly, n-doped films of TDPP-CNQx exhibit a good stability, retaining 85% of their initial conductivity (up to 62 S cm^{-1}) after 12 days under inert conditions, and maintaining a remarkable σ of $\sim 20 \text{ S cm}^{-1}$ after 6 h of air exposure (Figure S35). In fact, the stability of this polymer is superior to typical n-type materials, whose n-doped states rapidly degrade by oxygen and moisture (see Note S5). These results highlight the great potential of the CNQx building block for developing high-performance n-type OTE polymers, paving the way for practical applications.

Film Microstructures and Morphology. To further rationalize the above results, we investigated the impact of the structural modification on the polymer film morphology and microstructure, which was initially probed through atomic force microscopy (AFM) characterizations of the films prepared under the same conditions used for optimal device fabrication. As shown in Figures 6, S36 and S37, all Qx-based films exhibit finely interpenetrating networks with remarkably

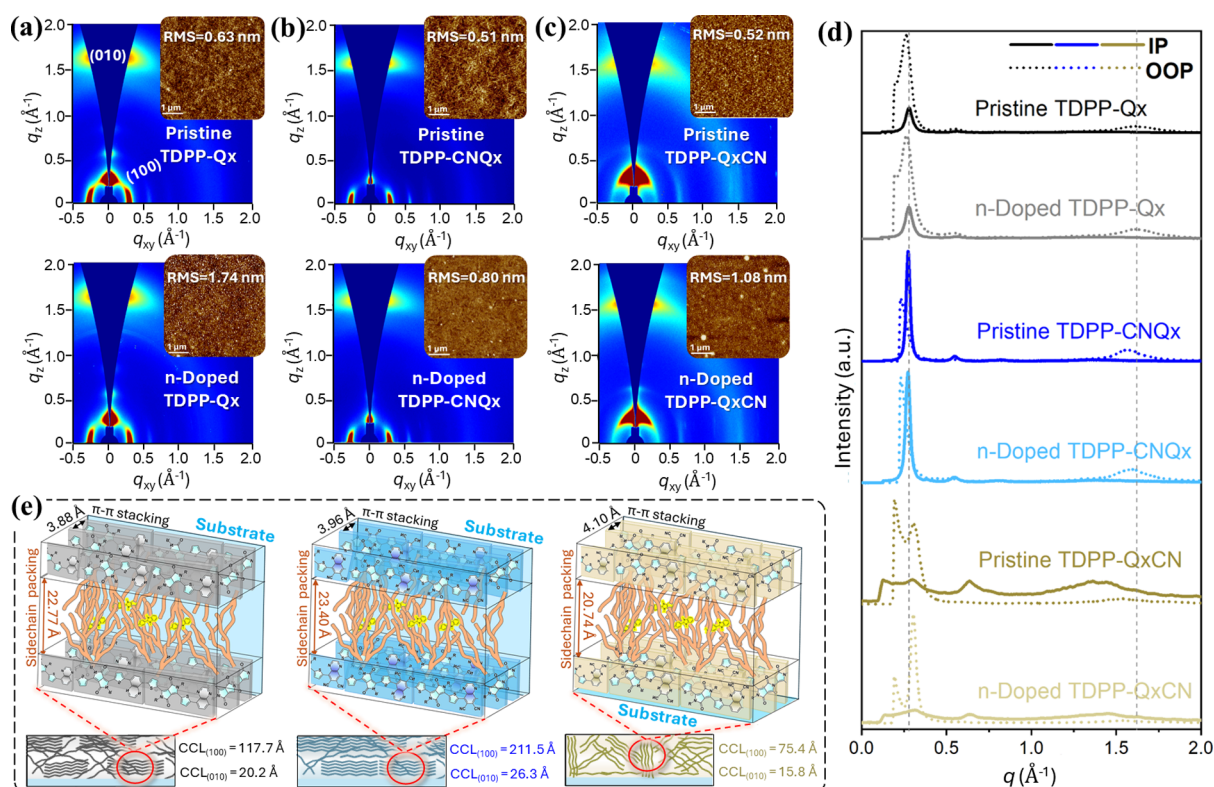


Figure 6. AFM height and 2D-GIWAXS images of (a) TDPP-Qx, (b) TDPP-CNQx and (c) TDPP-QxCN polymer films processed under the optimal device fabrication conditions. (d) In-plane (IP) and out-of-plane (OOP) line-cut profiles derived from the corresponding 2D-GIWAXS images (see Figure S38 for the unscaled raw line-cut data). (e) Schematic illustration of the effect of selective cyanation on the microstructures of the three polymers, with n-doped TDPP-Qx (gray) and TDPP-CNQx (blue) adopting predominantly face-on orientations, and n-doped TDPP-QxCN (gold) favoring an edge-on orientation. The JLBI molecular dopant is depicted in yellow.

smooth features, as evidenced by their low surface root-mean-square (RMS) roughness values of ~ 0.5 nm. However, TDPP-CNQx displays a more obvious fibrillar-like morphology, a feature that becomes even more pronounced after n-doping, potentially contributing to its superior thermoelectric performance. In this context, n-doping of all polymers with JLBI leads to slightly rougher film surfaces, with the RMS roughness increasing to 1.74 nm for TDPP-Qx, 0.80 nm for TDPP-CNQx and 1.08 nm for TDPP-QxCN. Interestingly, some obvious bright spots were observed in the height AFM images of n-doped films, presumably originating from the sequential doping strategy adopted where dopant molecules may aggregate on the film surface.⁷² These dopant-rich domains are found to be significantly smaller in the doped TDPP-CNQx films, reflecting an improved miscibility between JLBI and this polymer in accordance with its superior OTE performance. This enhanced miscibility is likely due to the more flexible backbone of TDPP-CNQx compared to the conformationally locked structures of its analogues, which should facilitate dopant diffusion into the polymer nanodomains and promote the formation of heavily doped phases.¹¹

To deepen our understanding of the supramolecular packing and molecular orientation of the Qx-based copolymers, we performed 2D grazing incidence wide-angle X-ray scattering (2D-GIWAXS) measurements on the corresponding thin films. Figure 6a–c display the 2D-GIWAXS patterns of TDPP-Qx, TDPP-CNQx and TDPP-QxCN respectively, with the associated line-cut profiles presented in Figures 6d and S38. As shown in Figure 6a, the control polymer TDPP-Qx adopts a predominant face-on orientation, as evidenced by a

pronounced (100) lamellar diffraction peak along the in-plane (IP) direction at $q_{xy} \approx 0.281 \text{ \AA}^{-1}$ and a well-defined (010) peak in the out-of-plane (OOP) direction at $q_z \approx 1.616 \text{ \AA}^{-1}$ arising from cofacial π - π stacking. These peaks correspond to lamellar and π - π stacking distances of 22.36 and 3.89 Å, respectively. Interestingly, introducing CN groups at the 6,7-positions of the Qx unit in TDPP-CNQx not only preserves the predominant face-on orientation but also significantly improves structural order, evidenced by the extension of lamellar diffraction up to the third order (300) in the IP direction and enhanced intensity in both (100) and (010) peaks, clear indicators of enhanced film crystallinity (Figure 6b). Nevertheless, TDPP-CNQx displays slightly larger lamellar and π - π stacking distances of 23.01 Å ($q_{xy} = 0.273 \text{ \AA}^{-1}$) and 4.01 Å ($q_z = 1.565 \text{ \AA}^{-1}$) compared to its noncyanated counterpart. These findings suggest that, although cyanation in TDPP-CNQx enhances crystallinity, it also introduces subtle increases in intermolecular spacing due to increased conformational flexibility or steric effects imparted by the CN groups. Unexpectedly, shifting the CN substitution to the 2,3-positions in TDPP-QxCN dramatically alters its molecular organization, resulting in a predominant edge-on orientation. This is manifested by hump-shaped (100) diffraction peaks in both IP and OOP directions at $q = 0.303 \text{ \AA}^{-1}$ (corresponding to a lamellar spacing of 20.74 Å), as well as a broadened (010) diffraction peak in the IP direction at $q_{xy} = 1.517 \text{ \AA}^{-1}$ (corresponding to a π - π stacking distance of 4.14 Å). In addition to a larger π - π stacking distance, it should be noted that TDPP-QxCN shows significantly broader and weaker diffraction features, indicating much lower degree

of structural ordering compared to its analogues (see Figure S38). To quantitatively evaluate crystallinity differences, the crystal coherence lengths (CCLs) from the (100) and (010) diffraction peaks were calculated according to the Scherrer equation, which quantifies the extent of crystalline order along specific directions.⁷³ As summarized in Table S5, the $CCL_{(100)}/CCL_{(010)}$ values progressively decrease from TDPP-CNQx (221.6/29.2 Å) to TDPP-Qx (124.5/20.9 Å), and ultimately to TDPP-QxCN (69.1/15.6 Å). This trend clearly underscores the critical role of regioselective cyanation in modulating polymer microstructure, enhancing crystallinity and intermolecular ordering when placed at the 6,7-positions, but disrupting chain organization when positioned at the 2,3-sites. We attribute the reduced crystallinity of TDPP-QxCN to its higher backbone rigidity, which may restrict molecular motion freedom and hinder the self-assembly process during film formation, ultimately compromising its charge transport properties.⁴⁷ In contrast, the most intense GIWAXS scattering features and largest CCLs observed for TDPP-CNQx reflect its superior intermolecular ordering, which accounts for its highest electron mobility measured in OTFTs.

Given that the incorporation of molecular dopants into conjugated polymer matrices is often associated with significant structural disruption, we also investigated how n-doping influences the microstructure of our materials by analyzing the GIWAXS patterns of polymer films doped with 3 mg mL⁻¹ of JLBI. Thus, it was found that the diffraction patterns remained nearly unchanged across all three polymers (bottom panels of Figure 6a–c), suggesting that their microstructural order was well-preserved even under high dopant loadings. However, while the π – π stacking distances were largely unaffected during the doping process, the lamellar spacing increases by up to 0.4 Å in JLBI-doped films of TDPP-Qx and TDPP-CNQx, with a smaller increase of just 0.1 Å observed in TDPP-QxCN (see Table S5). We tentatively interpret this behavior as indicative of partial intercalation of dopant molecules between the polymer's alkyl side chains (see schematic in Figure 6e), with the minimal effect observed in TDPP-QxCN likely attributed to its inherently more amorphous nature, which limits organized lamellar expansion.^{74–76} The preservation of crystallinity is further supported by the retained integrated areas of both (010) and (100) diffraction peaks in doped films of TDPP-Qx and TDPP-CNQx, exhibiting $CCL_{(100)}/CCL_{(010)}$ values of 117.7/20.2 Å and 211.5/26.3 Å, respectively. In contrast, JLBI-doped TDPP-QxCN films revealed slight increases of 6.2 and 0.3 Å compared to the neat film, reaching $CCL_{(100)}/CCL_{(010)}$ values of 75.4/15.8 Å. This expansion suggests that the low crystallinity of TDPP-QxCN may facilitate dopant diffusion through its amorphous regions, which can locally enhance ordering through dopant-induced intermolecular interactions but not to a level sufficient to enable efficient charge transport.^{41,68,77} In stark contrast, the coexistence of large and well-connected crystalline domains interspersed with optimally disordered phases in TDPP-CNQx, as revealed by Raman spectroscopy, creates a dual benefit: it promotes efficient electron transport through the crystalline network while facilitating rapid dopant diffusion through the amorphous regions.^{45,78,79} These synergistic effects account for the large n-doping efficiency observed in JLBI-doped TDPP-CNQx films and, when combined with the polymer's efficient charge transport pathways, give rise to its superior

electrical conductivity and enhanced thermoelectric performance.⁸⁰

CONCLUSIONS

In summary, we designed and synthesized two novel electron-deficient regioisomers (CNQx and QxCN) via selective cyanation of the quinoxaline unit. Importantly, these building blocks were synthesized via a straightforward, single-step procedure from low-cost precursors, offering a clear advantage in terms of scalability and practical applicability over conventional electron-deficient building blocks that often require laborious multistep synthesis. Their copolymerization with a thiophene-flanked diketopyrrolopyrrole (TDPP) yielded A–A type copolymers TDPP-CNQx and TDPP-QxCN, revealing that subtle differences in CN substitution strongly impact electronic structure and solid-state organization, ultimately governing charge transport and doping efficiency. Notably, cyanation at the 6,7-positions in TDPP-CNQx enables outstanding thermoelectric properties, with JLBI-doped films exhibiting a high σ of 44.50 S cm⁻¹ under ambient processing conditions and achieving a record thermoelectric figure of merit ($ZT = 0.20$) at room temperature. This performance surpasses all previously reported air-processed n-type organic materials and nearly all vacuum-processed counterparts (except one),⁷¹ firmly establishing CNQx as a promising design motif for the next-generation plastic electronics. Overall, this study advances the state-of-the-art in n-type thermoelectric materials by bridging synthetic accessibility, device performance, and air-processable compatibility.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c14816>.

Experimental methods, computational details, synthetic procedures for monomers and polymers, device fabrication, doping protocols, OTE characterization, morphological characterization data (AFM and 2D-GIWAXS) and Raman spectra are included (PDF)

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Notes

The authors declare no competing financial interest.

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